

Problem 3. Read Up:**3a) Read these pages from the course www site slowly, from beginning to end:****A1. Main Page**Found at this web address: <http://ruina.tam.cornell.edu/Courses/ME4730%20Fall%202017/index.html>As of October 14th, 2017, when you get there it should look like this:

The screenshot shows a web browser window with the address bar displaying <http://ruina.tam.cornell.edu/Courses/ME4730%20Fall%202017/index.html>. The page content is as follows:

MAE 4730/5730, Fall 2017:

Intermediate Dynamics and Vibrations

Enrollment is open. It may appear blocked because of a registrar error. That will be fixed. You CAN take the class.

This is a shell, duplicated from fall 2016. WWW pages subject to change. Most recent update on Aug 26, 2017.

Professor: [Andy Ruina, <ruina@cornell.edu>](mailto:ruina@cornell.edu)
TA/grader: Jared Foster [<jbf264@cornell.edu>](mailto:jbf264@cornell.edu)

This is the second in a three-course sequence. The first is *Dynamics* MAE 2030, the third is *Advanced Dynamics*, MAE 6700.
[More information.](#)
[ABET Syllabet](#)

Course communications use [Piazza](#): Please register, check for course news, and use Piazza to ask questions and make comments.

Course information

Lectures: MWF 11:15-12:05
Bard 140 (starting Wednesday Aug 23)
Office hours: Andy Ruina: see Andy's home page, [Andy Ruina](#)
Jared Foster: M & F 2 - 4:30, Thurston 102

[Syllabus](#)
[Homework policy](#)

Exams and grading: Grade determined by homeworks, project, prelims and final exam. All other things being exactly equal, 4730 students will tend to have higher grades.
Prelims: 1) Oct 17 and 2) Nov 14 (comprehensive).
Both Hollister B14, 7:30 - 9:00 (+ extra time).
Final: Thursday Dec 7, 2-4:30 PM, place TBD. Comprehensive. No notes. No calculators.

Main Resources

1. Background mechanics for this class: [Ruina & Pratap](#).
2. Textbooks: no official textbook, but if you like books, here are some options

Dynamics. Any book on Dynamics at a higher level than Ruina and Pratap (above), e.g. [Greenwood](#), [Synge&Griffith \(pdf\)](#), [O'Reilly](#), [Moon](#), [A. Rao](#), [Meirovitch](#), ... Probably

Carefully read all content on page.

A2. Homework Policy

Located at this web address: <http://ruina.tam.cornell.edu/Courses/ME4730%20Fall%202017/homework.html>

To get there from the homepage, click on the **boxed** link shown in the image below (click that link on the homepage, not the image):



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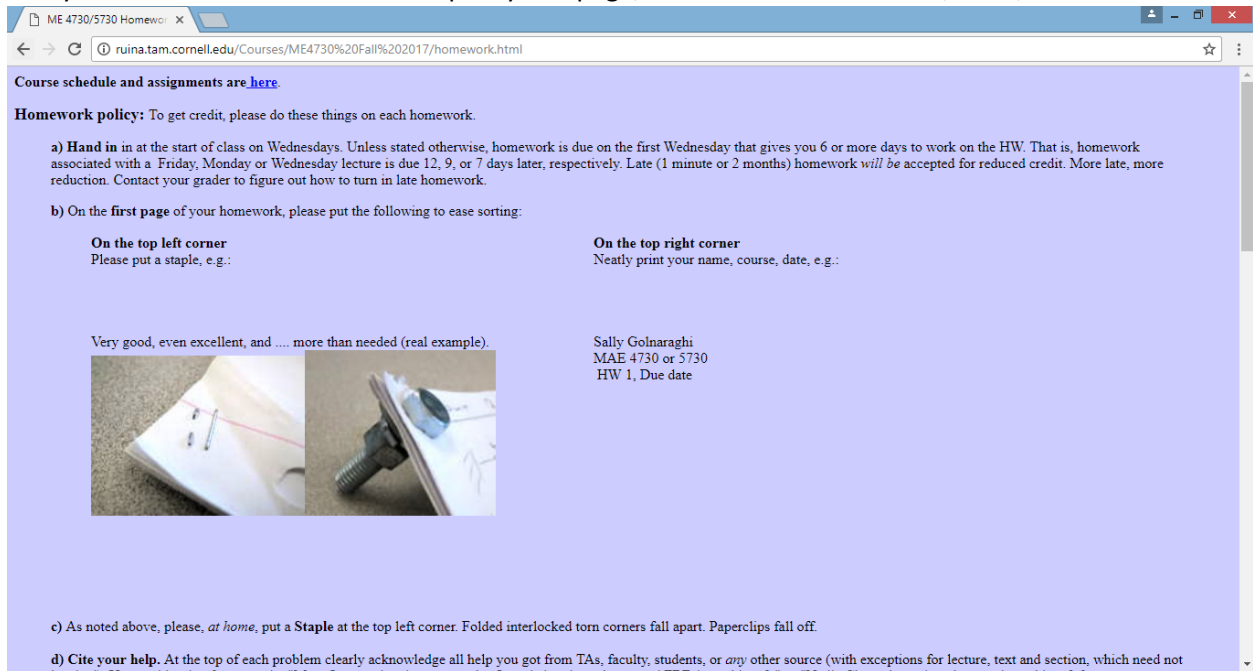
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And you will arrive at the homework policy webpage, where as of October 14th, 2017, it will look like this:



Course schedule and assignments are [here](#).

Homework policy: To get credit, please do these things on each homework.

- a) **Hand in** at the start of class on Wednesdays. Unless stated otherwise, homework is due on the first Wednesday that gives you 6 or more days to work on the HW. That is, homework associated with a Friday, Monday or Wednesday lecture is due 12, 9, or 7 days later, respectively. Late (1 minute or 2 months) homework *will be* accepted for reduced credit. More late, more reduction. Contact your grader to figure out how to turn in late homework.
- b) On the **first page** of your homework, please put the following to ease sorting:

On the top left corner Please put a staple, e.g.: Very good, even excellent, and more than needed (real example). 	On the top right corner Neatly print your name, course, date, e.g.: Sally Golnaraghi MAE 4730 or 5730 HW 1, Due date
--	--
- c) As noted above, please, *at home*, put a **Staple** at the top left corner. Folded interlocked torn corners fall apart. Paperclips fall off.
- d) **Cite your help.** At the top of each problem clearly acknowledge all help you got from TAs, faculty, students, or *any* other source (with exceptions for lecture, text and section, which need not

Carefully read all content on this page.

A3. Tentative Syllabus

Located at this web address:

<http://ruina.tam.cornell.edu/Courses/ME4730%20Fall%202017/SyllabusDetailed.html>

To get there from the homepage, click on the **boxed** link shown in the image below (click that link on the homepage, not the image):

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Syllabus (boxed link)
[Homework policy](#)
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And you will arrive at the tentative syllabus webpage, where as of October 14th, 2017, it will look like this:

Readings and Homework, Fall 2017
ME 4730/5730, Dynamics and Vibrations

---- SUBJECT TO CHANGE, all semester long ----
 ---- This page in progress on 8/26/2017 ----

Please read the [Homework policy](#) a few times, once before doing each of the first few [homework assignments](#).

Homework Solutions: TBD

Planned coverage (what you should know by the end of the semester)

Dynamics: Basic mechanics of 2D mechanisms by Newton-Euler, Lagrange or DAEs. Theory and practice

Basics: [Ruina & Pratap \(RP\)](#) The whole book (mostly review, but you should *master* all of it).

Topics which go a bit beyond many of the standard books

- 1) Using **Matlab** for simulations, plotting and animation.
- 2) Use of **Angular Momentum balance** to get equations of motion for complex systems.
- 3) **Mechanisms** (linked rigid objects)
- 4) **DAE** f (Differential Algebraic Equations) formulation of equations of motion

Vibrations: We will cover most of the content of most elementary Vibrations books.
 Forced, damped harmonic oscillator (inside and out).
 Linearization, normal modes and damped normal modes, matrix exponential, structural vibrations.

Lecture and homework schedule (Fall 2017, to be updated)

Homework is due the first Wednesday that is at least 6 days after the assigned date. It is subject to change until midnight on the end of Thursday 6 days before it is due.

1) August 23 Wed

Carefully read all content on this page.

A4. Also, try out each link that you think may be of use to you.

These include links on the homepage, links on the homework policy page, links on the syllabus page, and any links that can be accessed through the website.

Any conflicting or unclear information should be brought to the instructor's attention via a Piazza post (see part **3c** below).

3b) Read these sections of the Ruina/Pratap pdf book, available from Ruina's www page:

B1. Textbook can be accessed directly at this web address as of October 14th, 2017:

<http://ruina.tam.cornell.edu/Book/RuinaPratap-Jan-20-2015.pdf>

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Then click on the **boxed** link (shown in the image below) on the following page to access the textbook .pdf file:

Introduction to Statics and Dynamics
 Andy Ruina and Rudra Pratap
[Statics and Dynamics](#) 42 MB, 1065 pages (1/20/2015)
 Partial Solns Manual ([Feb 5, 2014](#))
[Errors](#) (you can edit this google doc)

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 for Oxford University Press
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Questions, comments, or suggestions? Send [email](#).

The story. In 1992 we told a textbook sales rep that we didn't like the Dynamics books. She said "Write your own." Most people said "Don't write a book. It's too much work, no-one will appreciate it and it will probably lose money." Saunders contracted the book and then sold it to Oxford. Now it's ours. Most years we work together for a month or so in Ithaca, Finland, or Bangalore. We re-organized the book 4 or 5 times and added a Statics book because students needed help with vectors and free body diagrams. Various pre-release versions have been used at Cornell and elsewhere about 25 times. It has had about 40 external reviews. The table of contents, preface and introduction show our intentions.

Use it for a course? Ask.

Help us:

- 1) Add to the solutions manual? We'll tell you how.
- 2) Suggestions are welcome: Errors, things that are confusing, things that are hard to find. Give us some clues about what you don't like.
- 3) **Make money** if you are the first to find an error: \$1.00 for math errors, \$5 for real errors (with suggestion about how to fix). How? List errors like [this](#) (if John Smith is

Or, obtain a hardcopy of the book.

B2. After accessing the pdf, read the following:

B2a. Chapter 1 (read)

(pg. 30-44 in the .pdf file, pg. 26-40 in the hardcopy book)

B2b. Chapter 2 (skim, only study things you don't know well already, make sure that within a few weeks you know all of this well.)

(pg. 46-129 in the .pdf file, pg. 42-125 in the hardcopy book)

B3c. Chapter 3 (read)

(pg. 130-193 in the .pdf file, pg. 126-189 in the hardcopy book)

B4d. Section 17.1 (read)

(pg. 869-883 in the .pdf file, pg. 865-879 in the hardcopy book)

B5e. Appendix A (read)

(pg. 1008-1019 in the .pdf file, pg. 1004-1015 in the hardcopy book)

3c) Write the following, if true, if not write that which is true:

"I have carefully done some of the reading assigned. That which I don't understand or agree with, I have posted on the course Piazza site. I haven't yet carefully read X [make appropriate substitutions for X]"

On your homework submission for this question, write whatever version of the sentence above is most true.

In the process of completing parts **3a** and **3b**, if there was something you didn't understand or agree with, make a Piazza post on the MAE 4730/5730 course page asking your question or voicing your disagreement.

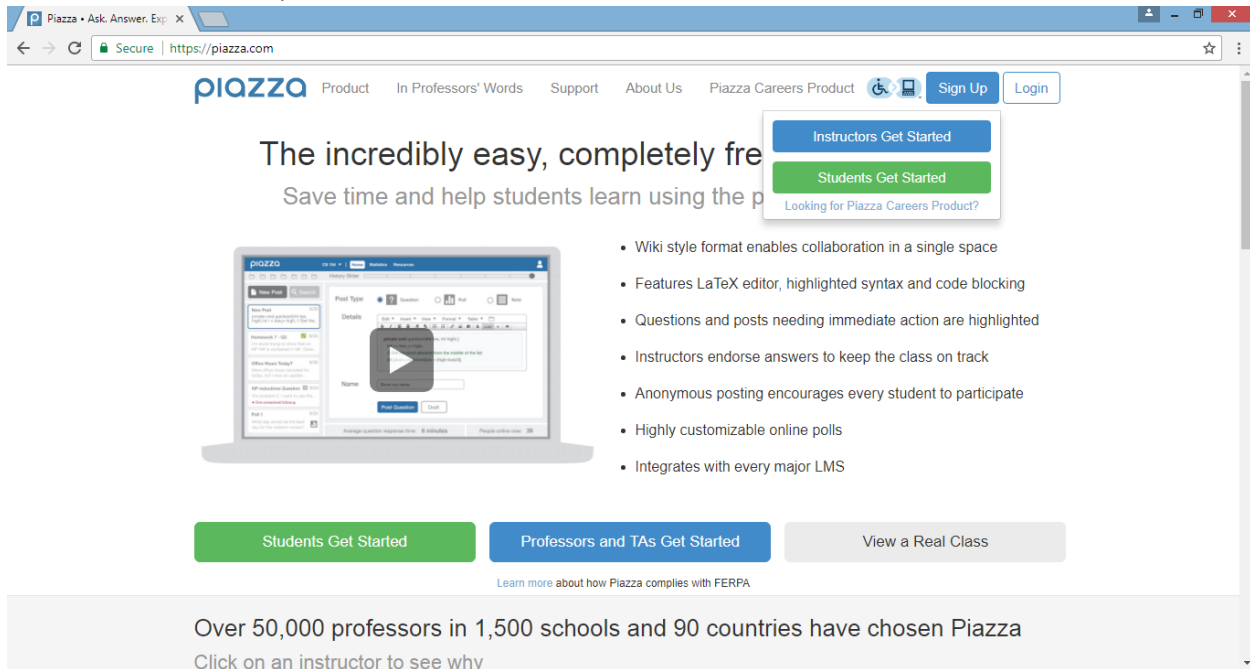
If you don't know how to do this, follow the following Piazza tutorial:

Piazza Class Sign-Up Tutorial:

(If you have already enrolled in the class on Piazza, log in, go to the MAE 4730/5730 Piazza course page, and skip to step 6, otherwise start at step 1 to enroll in the course.)

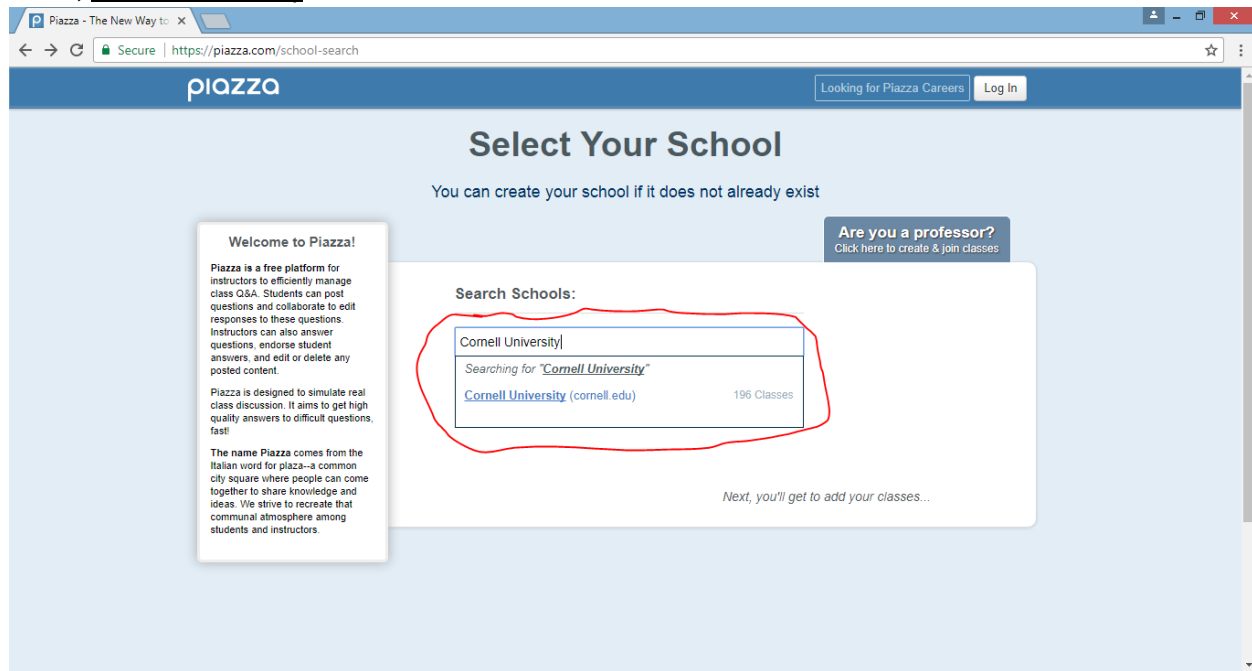
(NOTE: This tutorial is the same whether you already have a Piazza account or not.)

1. Go to <https://piazza.com/signup/>. Start by either clicking the "Sign Up" button at the top right of the webpage and then clicking the "Students Get Started" button in the window that appears, or by clicking the "Students Get Started" button directly underneath the welcome video:



This will bring you to the next page. Continue to step 2.

2. Type “Cornell University” (without quotes) into the Search Schools search-bar that appears and click on the top result, Cornell University:



This will bring you to the next page. Continue to step 3.

3. On the next page, type either “MAE 4730” (without quotes) or “MAE 5730” (without quotes) into any of the class search bars (1) and then click on the top result to select MAE 4730/5730: Intermediate Dynamics,

Welcome to Piazza!

Piazza is a free platform for instructors to efficiently manage class Q&A. Students can post questions and collaborate to edit responses to these questions. Instructors can also answer questions, endorse student answers, and edit or delete any posted content.

Piazza is designed to simulate real class discussion. It aims to get high quality answers to difficult questions, fast!

The name Piazza comes from the Italian word for piazza—a common city square where people can come together to share knowledge and ideas. We strive to recreate that communal atmosphere among students and instructors.

Cornell University
(change school)

Are you a professor?
Click here to create & join classes

Selected Term: Fall 2017

Fall 2017

Class 1: MAE 5730

Class 2: Searching for "MAE 5730"

Class 3: MAE 4730/5730: Intermediate Dynamics and Vibrations 72 Enrolled

Class 4:

Class 5:

Add Another Class

Join Classes

then check the box “Join as Student” (2), and a green check mark will appear by the “Join as”, confirming your class selection. After your selection is confirmed, click the “Join Classes” button (3) at the bottom:

Welcome to Piazza!

Piazza is a free platform for instructors to efficiently manage class Q&A. Students can post questions and collaborate to edit responses to these questions. Instructors can also answer questions, endorse student answers, and edit or delete any posted content.

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Cornell University
(change school)

Are you a professor?
Click here to create & join classes

Selected Term: Fall 2017

Fall 2017

Class 1: MAE 4730/5730: Intermediate Dynamics and Vibrations (edit)
Instructors: Andy Ruina - 72 Enrolled

Join as: ☒ Student ☐ TA ☐ Professor

Class 2:

Class 3:

Class 4:

Class 5:

Add Another Class

Join Classes

This will bring you to the next page. Continue to step 4.

4. On the next page, enter in your Cornell email and click on the “Submit Email” button:

The screenshot shows the Piazza website interface for Cornell University. At the top, there's a blue header with the 'piazza' logo and a 'Log In' button. Below the header, the page title is 'Cornell University' with a '(change school)' link. A 'Selected Term' dropdown is set to 'Fall 2017'. A list of classes is shown, with the first one being 'MAE 4730/5730: Intermediate Dynamics and Vibrations'. Below the class list, there's a section titled 'Please enter your school email address' with instructions to enter a Cornell.edu email. Two input fields are provided: 'Email:' and 'Confirm Email:', both containing 'YourEmail@cornell.edu'. A red circle highlights these fields and the 'Submit Email' button below them. At the bottom, there's a small link: 'Unable to sign up? Email us at team@piazza.com and we'll help you get started!'.

Based on whether or not a Piazza account already exists for that email address, the next screen will either have a log in box (follow step **a** below), or a box asking for the verification code that was sent to your email after clicking the “Submit Email” button (follow step **b** below)

- If you **do** already have a Piazza account, enter your password on the resulting screen to log in and confirm your enrollment.
- If you **do not** already have a Piazza account, follow the instructions on the resulting screen and create your account. Log in to proceed.

- After logging in, confirm you are on the MAE 4730/5730 Piazza page by checking to make sure it says MAE 4730/5730 at the top left of the screen on the blue bar, directly to the right of the Piazza logo. If it does not, click on whatever class it does say there and select MAE 4730/5730 from the resulting drop box.

Or, after logging in, follow this link to go directly to the MAE 4730/5730 Piazza page:

<https://piazza.com/class/j6nun4ob4yr3o8>

- You should now be on the MAE 4730/5730 Piazza page. Read through the posts. If you have a question or disagreement, click on the “New Post” button in the top left, type and post your question/disagreement. An example of a successfully posted disagreement will look something like the example below:

The screenshot shows a web browser window displaying a Piazza post. The browser's address bar shows the URL <https://piazza.com/class/j6nun4ob4yr3o8?cid=23>. The Piazza interface includes a top navigation bar with the Piazza logo, the course name 'MAE 4730/5730', and links for 'Q & A', 'Resources', and 'Statistics'. A search bar is also present. On the left side, there is a sidebar with a 'New Post' button and a list of recent posts, including 'Problem 6b', 'Disagreement with Homework Policy - ...', 'Dates on Course Page (for Question 3)', 'Intro to cats', 'Including Basic Matlab Code', 'WEEK 8/27 - 9/2', 'Canon Ball Path & Vector', and 'Office Hours'. The main content area displays a post titled 'Disagreement with Homework Policy - Starting Each Problem on a Fresh Page' by 'Hi Professor Ruina'. The post text reads: 'In order to save paper, I would like to request that we are not required to start each homework problem on a fresh page. Best, [redacted]'. Below the post, there is a section for 'followup discussions' with a 'Resolved' status. A comment from 'Andy Ruina' dated '1 month ago' states: 'If you make it clear and easy to read you can start on the same page and use the backs of pages.' Another comment from '[redacted]' dated '1 month ago' says 'Thanks!'. At the bottom, a summary bar shows 'Average Response Time: 28 min', 'Special Mentions: Andy Ruina answered Q21 Matrix Singularity Issue in 7 min. 6 days ago', and 'Online Now: 6 | This Week: 66'.